Part 1 (Real Data Fitting): Focused on the AAV variant experimental data. Each of the 5 algorithms (CMA-ES, L-SHADE, PSO, Dual Annealing, Basin-Hopping) was run 13 independent times, with each run budgeted at ~21,000 function evaluations (SSE calculations). From these, the single best-fitting parameter set (lowest SSE) per algorithm was selected as its "ground truth" parameters, stored in files like data/*_best_params.csv.
Part 2 (Synthetic Data Benchmarking): Using those 5 ground truth parameter sets, you generated 3 synthetic datasets per set (total 15 datasets, e.g., cmaes_1.csv, etc.), each with realistic noise added via the composite model. Then, each algorithm was benchmarked by fitting all 15 datasets, with 10 independent runs per algorithm-dataset pair (5 algos × 15 datasets × 10 runs = 750 total runs). Full run histories are available for these (and for Part 1), including per-evaluation details: parameter vectors, SSE values, and timestamps.

Introduction

Parameter estimation for nonlinear dynamic models remains a fundamental challenge in systems biology and synthetic biology, where complex mechanistic models often exhibit ill-conditioned, multi-modal error surfaces. Traditional gradient-based optimizers frequently converge to suboptimal local minima, particularly when fitting high-dimensional models to time-course data from gene regulatory circuits. A representative example is the tetracycline repressor (TetR)-controlled green fluorescent protein (GFP) expression system, modeled as a system of ordinary differential equations (ODEs) with 11 state variables (including TetR, inducer anhydrotetracycline (aTc), mRNA, GFP, and intermediates like small transcription activating RNA (STAR) and toehold switch (THS)) and 21 kinetic parameters governing transcription, translation, binding, degradation, and other processes [1,2]. Prior research has demonstrated that stochastic global optimization methods, such as evolutionary algorithms, are essential for reliable parameter recovery in such scenarios [3,4].

Several global optimization algorithms have shown promise for these tasks. Covariance Matrix Adaptation Evolution Strategy (CMA-ES) is a derivative-free method effective for non-convex, noisy objectives [5]. L-SHADE, an adaptive differential evolution variant, excels in high-dimensional problems through success-history adaptation and linear population size reduction [6]. Particle Swarm Optimization (PSO), inspired by social behavior, balances exploration and exploitation [7]. Dual Annealing combines simulated annealing with local search for global minima [8], while Basin-Hopping employs multi-start perturbations with gradient-based refinement [9]. These algorithms represent diverse stochastic strategies, enabling a comparative evaluation.

This study focuses on benchmarking these five optimizers (CMA-ES, L-SHADE, PSO, Dual Annealing, and Basin-Hopping) on calibrating the TetR-GFP model. To establish ground truth, each algorithm was initially run 13 times to fit experimental AAV variant data, with the best result selected as the ground truth parameter set for that algorithm. Synthetic datasets were then generated from these ground truth sets using a realistic composite noise model (incorporating multiplicative fluctuations, heteroscedastic additive noise, autocorrelation, and linear drift) to mimic experimental artifacts in AAV-like data. The optimizers were assessed on recovering these ground truth parameters from the synthetic datasets, all simulated under the AAV variant conditions. We hypothesize that adaptive evolutionary methods (e.g., CMA-ES and L-SHADE) will outperform others in accuracy, consistency, and robustness to noise due to the model's complexity. The significance lies in providing empirical guidance for optimizer selection in biological modeling, highlighting the impact of noise realism on evaluation, and promoting efficient parameter estimation workflows for synthetic gene circuits.

Methods

Model Description

The study utilizes a mechanistic ODE model of a TetR-regulated RNA-protein hybrid gene circuit expressing GFP, implemented in Python (protein_model.py) and MATLAB (Protein_Model.m). The model consists of 11 state variables, including concentrations of STAR, THS, TetR, aTc, TetR-aTc complex, mRNA, and GFP, evolving via coupled ODEs with 21 parameters (e.g., transcription/translation rates, binding/unbinding constants, degradation rates). Simulations use SciPy's solve_ivp with the BDF solver, over 420 minutes (7 hours) at 0.1-second steps, with tolerances of 1e-11. Initial conditions set all states to zero except aTc at ~1e-7 M (50 units) for induction at t=0.

Experimental data for the AAV variant (with TetR degradation tag multiplied by 2) was used, measured every 10 minutes in a plate reader, averaged across three biological replicates, and provided as mean traces in .mat files loaded as NumPy arrays. Other variants (ASV, LVA, noTetR) are defined in the model but not fitted; all fitting and simulations focused on AAV conditions.

Ground Truth Parameter Estimation

To obtain ground truth parameter sets, each of the five algorithms was run 13 times on the experimental AAV data, minimizing the sum of squared errors (SSE) between simulated GFP and the mean experimental trace (objective_function.py). The best (lowest SSE) result from these 13 runs per algorithm was selected as the ground truth parameter set (stored in data/*_best_params.csv).

Synthetic Data Generation and Noise Model

For each of the five ground truth parameter sets, a noise-free GFP trajectory was simulated under AAV conditions (simulate.py).

A composite noise model (noise_model.py, detailed in noise_model_report.md) was applied: (1) multiplicative noise $\delta_t \sim \mathrm{Normal}(0, 0.1^2)$ for proportional fluctuations; (2) heteroscedastic additive noise $\eta_t \sim \mathrm{Normal}(0, a + b \cdot x_t)$ with a=25, b=1.0 for signal-dependent variance; (3) autocorrelated error $\varepsilon_t = \rho \cdot \varepsilon_{t-1} + \eta_t$ with ρ=0.007 for temporal correlations; (4) linear drift $drift_t = 0.01 \cdot t$ (per minute). Noisy trace: $y_t = (1 + \delta_t) \cdot x_t + \varepsilon_t + drift_t$.

For each ground truth set, three independent datasets were created (generate_data.py), each with three replicate traces sampled every 10 minutes, yielding 15 CSV files (e.g., cmaes_1.csv), all resembling AAV experimental data.

Optimization Setup and Benchmarking

Optimizers minimized SSE with a penalty (1e12) for unstable simulations. Fits used AAV variant conditions with bounds on parameters (e.g., rates 5e-4 to 5e-1 min^{-1}, listed in fit_synthetic.py) spanning physiological ranges.

Algorithms: CMA-ES (cma package), L-SHADE (mealpy), PSO (pyswarms, 50 particles, c1=c2=1.49445, w=0.729), Dual Annealing (scipy.optimize.dual_annealing), Basin-Hopping (scipy.optimize.basinhopping with L-BFGS-B). Wrappers in optimizers.py.

Each optimizer ran on all 15 synthetic datasets with approximately 21,000 function evaluations (calculated as 21 parameters × 1,000 evaluations per parameter), N=10 independent runs per combination (different seeds), on HPC via SLURM scripts (*.sh). Best parameters and SSE saved; results aggregated (analyze_results.py) for metrics like RMSE and R^2, visualized with pandas, numpy, seaborn, matplotlib (plot_results.py, requirements.txt).

This setup ensures reproducibility, attributing differences to algorithm efficacy under realistic noise. All code is in the repository.